

NAMAN YESHWANTH KUMAR

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SUMMARY

MS Computer Engineering (ASU, GPA 3.81, May 2026) with end-to-end autonomous-driving stack experience across perception, fusion, localization, planning, and control. Implemented the full self-driving pipeline in C++ from scratch on the **Waymo Open Dataset** and **CARLA** — Extended Kalman Filter sensor fusion, LiDAR localization via ICP/SVD with **0.299 m pose error over 172 m (4× better than baseline, “outstanding” reviewer feedback)**, FSM behavior planner with cubic spiral trajectories, and PID control. Also built a physical autonomous car as a hardware capstone and trained visuomotor robot policies in MuJoCo (PyTorch + Diffusion Policy).

TECHNICAL SKILLS

C++ & AV Stack: C++ | CMake | Eigen | OpenCV | PCL (Point Cloud Library) | CARLA Simulator | MuJoCo | Real-Time Systems | I2C / Embedded | Python | Linux | Git

Perception, Fusion & Localization: LiDAR Point Cloud Processing | Bird’s-Eye View Projection | 3D Object Detection | Extended Kalman Filter | Multi-Object Tracking | ICP / SVD (from scratch) | NDT | Scan Matching | YOLOv4 | SSD ResNet18/50 | Waymo Open Dataset

Planning, Control & ML: Behavior FSM | Motion Planning | Cubic Spiral Trajectories | PID Control | PyTorch | TensorFlow | Diffusion Policy | Visuomotor Policy Learning

EDUCATION

Master of Science — Computer Engineering (GPA 3.81/4.0)

Arizona State University

Expected May 2026

Tempe, AZ

Coursework: Perception in Robotics · Reinforcement Learning in Robotics · Artificial Intelligence · Foundations of Algorithms · Algorithm/Hardware Co-Design

Bachelor of Engineering — Electrical and Electronics

BMS College of Engineering

August 2022

Bengaluru, India

PROJECT EXPERIENCE

End-to-End Self-Driving Pipeline — C++, Waymo Open Dataset, CARLA Simulator

Feb 2023

- **Perception & Sensor Fusion:** Detected vehicles in 3D LiDAR point clouds via bird’s-eye view projection using YOLOv4 (Darknet) and FPN ResNet18 on the Waymo Open Dataset; implemented an Extended Kalman Filter from scratch fusing LiDAR and camera measurements — full predict/update cycle, Mahalanobis distance data association, and track lifecycle (initialization, confirmation, deletion); validated via RMSE across multi-frame sequences.
- **Mapping & Localization:** Implemented ICP from scratch using SVD decomposition (KdTree nearest-neighbor association, centroid alignment, iterative convergence) and benchmarked against NDT and PCL-based ICP; achieved **0.299 m pose error over 172 m traveled** (requirement: <1.2 m) — **4× better than baseline**, earning “outstanding” reviewer feedback.
- **Planning & Control:** Built an FSM behavior planner (lane follow, nudge, lane change, intersection stop) with a motion planner generating cubic spiral trajectories evaluated by collision and lane-deviation cost functions; implemented PID controllers from scratch for steering (cross-track error) and throttle in CARLA simulator.

Autonomous Car — Physical Hardware Capstone (Raspberry Pi + Arduino)

Jan – Jun 2022

- Built a full physical autonomous car as bachelor’s capstone — OpenCV lane detection driving a PID steering controller, cascade classifiers for real-time traffic sign and obstacle detection (stop, U-turn, avoidance), Bluetooth remote control for manual data collection; Arduino motor driver controlled via I2C from Raspberry Pi, Python perception feeding C++ control logic.

Diffusion Policy — Visuomotor Policy Learning in MuJoCo Simulation

Apr 2025

- Adapted the Columbia Diffusion Policy architecture (RSS 2023) for learning robot control policies from demonstrations — configured MuJoCo simulation environments, ran end-to-end PyTorch training and rollout evaluation; team project in CSE598 Perception in Robotics.

SiliconCrew — Autonomous LLM Agent for Code Generation & Verification

Nov 2025 – Present

- Built a LangGraph ReAct agent (21 tools) with multi-provider LLM routing (Gemini/OpenAI/Anthropic), autonomous self-correction loop (failure → diagnose → patch → re-verify with zero human input), and a full MCP server (stdio/SSE/HTTP); achieved 37/92 first-pass and 5/5 on structured code generation benchmarks.

Trezzit — Production AI SaaS (trezzit.com)

Feb 2025 – Present

- Sole-built and shipped a production AI SaaS serving **600+ users** — two-pass LLM receipt pipeline with provider-abstracted registry across Gemini, OpenAI, Claude, and Grok; GitHub Actions CI with 995 automated tests; live at trezzit.com.

WORK EXPERIENCE

Systems Engineer

NCR GOLD

Oct 2022 – Apr 2024

Bengaluru, India

- Automated the entire order lifecycle for a wholesale business — production Django platform (PostgreSQL + Celery + Nginx) processing **18,000+ orders over 3 years**; reverse-engineered third-party ERP (zero public API) via SQL schema analysis for bidirectional sync, eliminating ~20 min of manual relay per order.

AWARDS & HONORS

- Ranked **Top 10 nationally** in e-Yantra Robotics Competition (IIT Bombay) — designed a self-balancing vehicle in CoppeliaSim, February 2022.
- Won **‘Best Use of AI’** at Strategy X DevHacks’25 (ASU) — built AdaptED AI, a Gemini-powered platform generating personalized flashcards, quizzes, and learning paths from uploaded documents in a 24-hour hackathon.
- Won **1st place at Devlabs Hackathon** — HEAL.AI, a healthcare AI assistant with a custom RAG pipeline (768-dim embeddings, cosine similarity retrieval) for insurance Q&A and medical bill error detection.